

Capable of measuring sparkle,
color sparkle, flop index,
and graininess

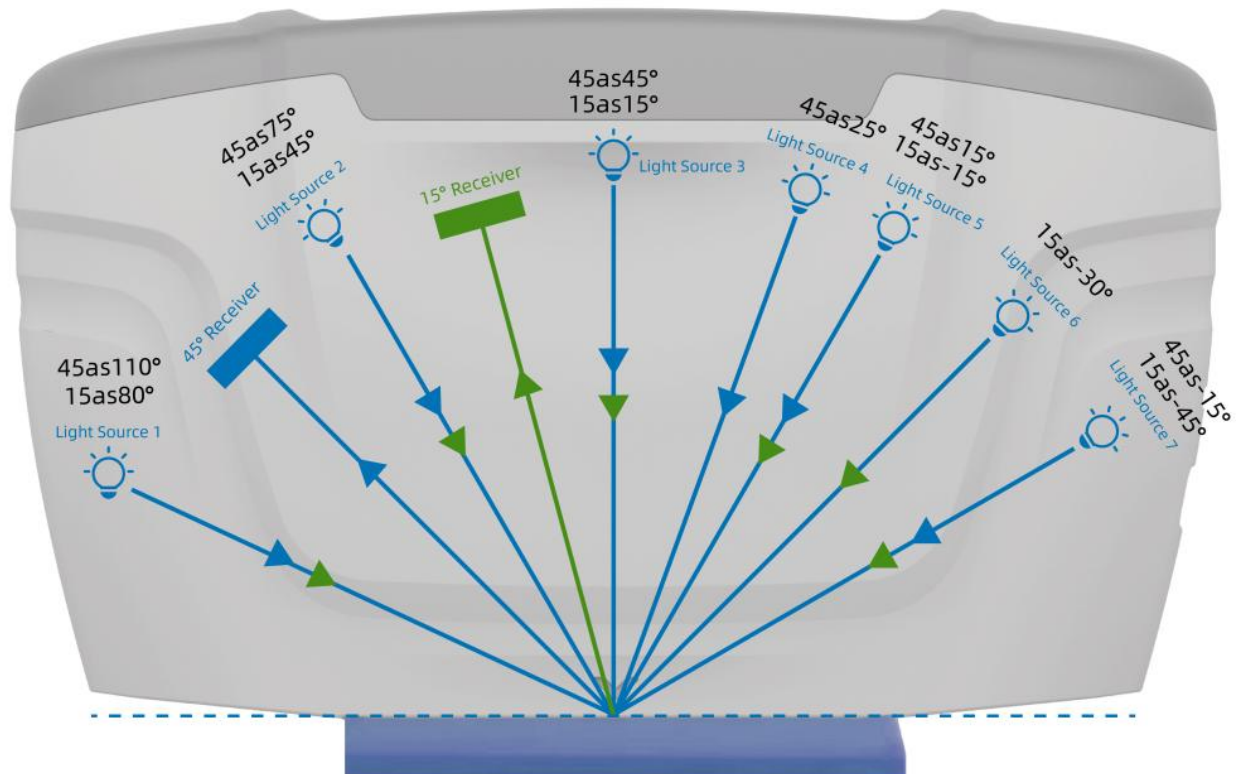
Multi-Angle Portable Spectrophotometer
3-Angle / 6-Angle / 12-Angle Color Measurement



1. Product Features

- Full-scene coverage: Up to 12 angles for accurate capture of automotive, plastic, cosmetics, and other effect finishes
- Four effect-parameter measurement system, ending the era of purely subjective evaluation
“subjective evaluation” in effect finish quality control
- 10-year / 5-million measurements light source revolution, setting a benchmark for industrial-grade reliability
- Outstanding 0.18 inter-instrument agreement, the “unified data language” for globalized production

2. Measurement principle



3. Technical Parameters

Model	TR 5402/03	TR 5402/06	TR 5402/12
Measurement Geometry	3 angles (3 light sources, 1 receiver)	6 angles (6 light sources, 1 receiver)	12 angles (7 light sources, 2 receivers)
Color Measurement Angles	45° receiver: 45as15°, 45as45°, 45as110°	45° receiver: 45as-15°, 45as15°, 45as25°, 45as45°, 45as75°, 45as110°	45° receiver: 45as-15°, 45as15°, 45as25°, 45as45°, 45as75°, 45as110° 15° receiver: 15as-45°, 15as-15°, 15as15°, 15as-30°, 15as45°, 15as80°
Illumination Source	Full-spectrum LED with blue-light enhancement		
Light Source Lifetime	10 years / 5 million measurements		
Detector	256-pixel dual-array CMOS image sensor		



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Spectral Range	400nm-700nm		
Measurement Aperture	Illumination Φ24mm / Sampling 9×12mm		
Additional Indices	Flop Index, Color Travel Index		
Measurement Time*	About 2.5s	About 4.5s	About 5s
Battery Life	Up to 5000 measurements		
Color Repeatability	0.02 ΔE*ab (after warm-up calibration, 30 white tile measurements at 5s intervals, standard deviation)		
Color Reproducibility	ΔE* < 0.10 (average on gray BCRA tiles); ΔE* < 0.25 (average on colored BCRA tiles)		
Inter-instrument Agreement	0.2 ΔE*00 (BCRA II, 12 tiles)		0.18 ΔE*00 (BCRA II, 12 tiles)
Effect Parameters	/	Sparkle, Color Sparkle, Graininess	
Effect Measurement	/	6-angle sparkle and graininess: 15as-45°, 15as-30°, 15as-15°, 15as15°, 15as45°, 15as80°; 15d diffuse graininess	
Effect Repeatability	/	Sparkle short-term repeatability: 0.12% (standard deviation of 10 measurements) (After instrument warm-up calibration, average of 10 measurements on color tile at 10s intervals) Graininess short-term repeatability: 0.09% (standard deviation of 10 measurements) (After instrument warm-up calibration, average of 10 measurements on color tile at 10s intervals)	
Effect Reproducibility	/	Sparkle reproducibility: 1.9% (standard deviation of 10 measurements) (Average value from measurements on 12 BCRA Series II color tiles) Graininess reproducibility: 1.4% (standard deviation of 10 measurements) (Average value from measurements on 12 BCRA Series II color tiles)	
Positioning	Color camera preview		
Display	3.5-inch TFT true-color capacitive touchscreen		
Data Storage	10,000 records		
Compliance Standards	ASTM D 2244,E 308,E 1164,E 2194,E2539,DIN 5033,5036,6174,6175-1,6175-2;ISO7724,11664-4 , SAE J 1545		
Special Features	Provides precise and consistent color measurement for metallic, pearlescent, and other complex special-effect colors		
Spectral Dispersion	Concave grating		
Wavelength Interval	10nm		
Measurement Range	0~600%		



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Half Bandwidth	10nm
Color Spaces	CIE LAB,XYZ,Yxy,LCh,βxy,DIN Lab99,RGB,hunter Lab
Color Difference Formulas	ΔEab, ΔE94, ΔEcmc(2:1), ΔEcmc(1:1), ΔE00, DIN ΔE99, ΔE DIN6175, multiple automotive OEM formulas; Metamerism, ΔEab(Hunter), ΔEch, ΔE*uv
Observer Angles	2°/10°
Observation Light Sources	A,B,C,D50,D55,D65,D75,F1,F2,F3,F4,F5,F6,F7,F8,F9,F10,F11,F12,CWF,U30,U35,DLF,NBF,TL83,TL84, ID50,ID65,LED-B1,LED-B2,LED-B3,LED-B4,LED-B5,LED-BH1,LED-RGB1,LED-V1,LED-V2,LED-8
Display Functions	Spectral graph/data, sample color values, color difference values/graph, pass/fail results, color simulation, color bias, effect values, effect difference values
Trigger Modes	Pressure sensor, button trigger, software trigger
Measurement Modes	Single, average, continuous measurement
Dimensions	194 × 73 × 118mm
Weight	About 850g
Interfaces	USB, Bluetooth
Languages	Simplified Chinese, Traditional Chinese, English
Operating Conditions	10°C–50°C, ≤85% RH, non-condensing
Storage Conditions	-20°C–50°C, ≤85% RH, non-condensing
Calibration	Built-in white reference, external white tile, black cavity, external color tile
Calibration Interval	4h, 8h, 24h, or at startup
Standard Accessories	Power adapter, data cable, user manual, quality management software (download), calibration kit, protective cap, wrist strap
Optional Accessories	Mini printer

*Color measurement time

**Technical specifications are for reference only and are subject to the actual product sold